Logarithmic Functions & Properties

ANSWER KEY

Section 1: Definition of Logarithms and Inverses

1 Evaluate $\log_4\left(\frac{1}{64}\right)$.

a. -3 ✓

b. 3

c. $-\frac{1}{3}$

d. 16

Let $y = \log_4(1/64) \Leftrightarrow 4^y = \frac{1}{64} = 4^{-3} \Rightarrow y = -3$.

2 What is the domain of $f(x) = \ln(5 - 2x)$?

a. $(-\infty, 2.5)$ ✓

b. $(2.5, \infty)$

c. $(-\infty, 5)$

d. $(0, \infty)$

The argument of a logarithm must be strictly positive: $5 - 2x > 0 \Leftrightarrow 2x < 5 \Leftrightarrow x < 2.5$. The domain is $(-\infty, 2.5)$.

3 Find the inverse function of $g(x) = 3 + 2\ln(x - 1)$.

a. $g^{-1}(x) = e^{(x-3)/2} + 1$ ✓

b. $g^{-1}(x) = e^{(x-3)/2} - 1$

c. $g^{-1}(x) = 2e^{x-3} + 1$

d. $g^{-1}(x) = e^{2(x-3)} + 1$

Set $y = 3 + 2\ln(x - 1) \Rightarrow y - 3 = 2\ln(x - 1) \Rightarrow \ln(x - 1) = \frac{y-3}{2} \Rightarrow x - 1 = e^{(y-3)/2} \Rightarrow x = e^{(y-3)/2} + 1$.

Section 2: Logarithmic Properties & Condensing Expressions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Express $2\log x - 3\log y + \frac{1}{2}\log z$ as a single logarithm.

a. $\log\left(\frac{x^2\sqrt{z}}{y^3}\right)$ ✓

b. $\log\left(\frac{2x\sqrt{z}}{3y}\right)$

c. $\log\left(x^2 - y^3 + \sqrt{z}\right)$

d. $\log\left(\frac{x^2}{y^3\sqrt{z}}\right)$

Power rule gives $\log(x^2) - \log(y^3) + \log(z^{1/2})$. Quotient and product rules combine to $\log\left(\frac{x^2\sqrt{z}}{y^3}\right)$.

5 If $\log_b 2 = 0.35$ and $\log_b 3 = 0.55$, what is the value of $\log_b 18$?

a. 1.45 ✓

b. 0.90

c. 1.25

d. 1.10

Note $18 = 2 \cdot 3^2$. By log properties: $\log_b 18 = \log_b 2 + 2\log_b 3 = 0.35 + 2(0.55) = 0.35 + 1.10 = 1.45$.

6 Using the change of base formula, express $\log_5 12$ in terms of natural logarithms.

a. $\frac{\ln 12}{\ln 5}$ ✓

b. $\frac{\ln 5}{\ln 12}$

c. $\ln 12 - \ln 5$

d. $\ln\left(\frac{12}{5}\right)$

The change of base formula states $\log_b a = \frac{\log_c a}{\log_c b}$. For base e, $\log_5 12 = \frac{\ln 12}{\ln 5}$.

Section 3: Solving Exponential and Logarithmic Equations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Solve for x: $3^{2x-1} = 27$.

a. 2 ✓

b. 1

c. 3

d. $\frac{3}{2}$

$27 = 3^3$. Equating exponents: $2x - 1 = 3 \Rightarrow 2x = 4 \Rightarrow x = 2$.

8 Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. 4 ✓

b. 4 and -2

c. 2

d. 8

Combine: $\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since the domain requires $x > 2$, $x = -2$ is extraneous. The only valid solution is $x = 4$.

9 Solve for t: $50e^{0.04t} = 200$.

a. $\frac{\ln 4}{0.04} \approx 34.66$ ✓

b. $\frac{\ln 150}{0.04}$

c. $\frac{4}{0.04} = 100$

d. $\ln(4 \times 0.04)$

Divide by 50: $e^{0.04t} = 4$. Take natural log: $0.04t = \ln 4 \Rightarrow t = \frac{\ln 4}{0.04} \approx 34.66$.

Section 4: Logarithmic Scales and Applications

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 The pH of a chemical solution is given by $\text{pH} = -\log[\text{H}^+]$. If the hydrogen ion concentration is 1.0×10^{-5} M, what is the pH?

a. 5.0 ✓

b. -5.0

c. 1.0

d. 9.0

$\text{pH} = -\log(10^{-5}) = -(-5) = 5.0$.

11 An earthquake measures 7.2 on the Richter scale, while a second measures 5.2. How many times more intense is the first earthquake?

a. 100 ✓

b. 2

c. 20

d. 1000

The Richter scale is logarithmic base 10. The difference in magnitudes is $7.2 - 5.2 = 2$. The intensity ratio is $10^2 = 100$ times greater.

12 What is the vertical asymptote of the graph of $f(x) = \log_3(x + 4) - 2$?

a. $x = -4$ ✓

b. $x = 4$

c. $y = -2$

d. $x = 2$

The vertical asymptote occurs where the argument equals zero: $x + 4 = 0 \Rightarrow x = -4$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647